

Semantic Collision in Scientific Retrieval: Verifying Insufficient Evidence for Biological Aging Queries Using Particle Physics and Computer Vision Literature

Input Question

Can we stop ourselves from aging?

Domain

Biology

Validation Status

needs_data

Problem Statement

The inquiry seeks to determine if biological aging can be halted, referencing a potential 'mortality plateau' at age 105+. However, the provided evidence set consists exclusively of literature on particle physics (stop squarks) and computer vision (stop sign detection), creating a semantic collision with the biological concept of 'stopping' aging. The core problem is the complete absence of domain-relevant evidence to support or refute claims about human senescence or mortality dynamics.

Rationale

Scientific rigor requires that all factual claims be grounded in provided evidence. Since the allowed evidence IDs (EV-Q025-8a6346824f77edf2b35a2b74, EV-Q025-b0f8d69a416d10c7427609b9, EV-Q025-7654cac739beaaaa3d5b8e33, EV-Q025-5cebb360904beebdc6f4d5dc) discuss supersymmetric particles and object detection algorithms rather than gerontology, no biological hypothesis can be validated. The research plan therefore focuses on formally verifying this insufficiency through semantic and keyword analysis, preventing the hallucination of biological mechanisms from irrelevant physical science data.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: Insufficient Evidence: No valid biological hypothesis regarding stopping human aging can be generated from the provided evidence cards, as they exclusively pertain to particle physics (stop squarks) and computer vision (stop sign detection).
- **Mechanism**: The semantic collision between the query term 'stop' (biological cessation) and the evidence content ('stop' as supersymmetric particle or traffic signal) renders the evidence set null for hypothesis generation. No biological mechanism exists in the allowed sources.

- ****Falsifiable Prediction****: If a comprehensive review of the four allowed evidence IDs (EV-Q025-8a6346824f77edf2b35a2b74, EV-Q025-b0f8d69a416d10c7427609b9, EV-Q025-7654cac739beaaaa3d5b8e33, EV-Q025-5cebb360904beebdc6f4d5dc) reveals any mention of human physiology, senescence, or mortality plateaus, this claim of insufficiency is falsified.
- ****Required Observations****: Verification that EV-Q025-8a6346824f77edf2b35a2b74 discusses stop squark mass splitting, not aging. ; Verification that EV-Q025-b0f8d69a416d10c7427609b9 discusses YOLO vs Faster RCNN for traffic signs, not biology. ; Verification that EV-Q025-7654cac739beaaaa3d5b8e33 discusses LHC cross-section limits, not gerontology. ; Verification that EV-Q025-5cebb360904beebdc6f4d5dc discusses SUSY parameter space, not centenarian mortality.
- ****Risk of Being Wrong****: Low risk. The quoted text in all evidence cards explicitly references physics and computer vision terminology unrelated to biology. The only risk is if the full text contains hidden biological data not reflected in the verified quotes, which is unlikely given the specific titles and locators.

Hypothesis 2

- ****Hypothesis****: Pending Validation Required: The 'late-life mortality plateau' at age 105 mentioned in the booklet excerpt remains an unverified knowledge gap because no allowed evidence ID supports or refutes demographic claims about supercentenarians.
- ****Mechanism****: Hypothesis generation is blocked by the absence of relevant evidence. The booklet excerpt asserts a 2018 Science paper finding, but strict adherence to allowed_evidence_ids prevents incorporating this as established fact. Therefore, the hypothesis is merely that 'current evidence is insufficient to evaluate the mortality plateau'.
- ****Falsifiable Prediction****: This hypothesis is weakened if any of the allowed evidence IDs are found to contain demographic data or citations to the 2018 Science paper on mortality plateaus.
- ****Required Observations****: Confirmation that none of the allowed evidence IDs contain keywords: 'mortality', 'centenarian', 'aging', 'senescence', 'demography'. ; Confirmation that the relevance scores (0.55-1.0) assigned to physics/vision papers were due to keyword matching on 'stop' rather than topical relevance.
- ****Risk of Being Wrong****: Moderate. While the quoted texts are clearly irrelevant, automated extraction systems sometimes misclassify documents. However, manual inspection of quotes confirms domain mismatch.

Technical Details

This experiment design addresses a 'mechanism_discovery' query regarding human aging by formally validating the 'Insufficient Evidence' hypothesis. The core technical challenge is not biological modeling, but rather rigorous domain verification and semantic gap analysis. The protocol involves: 1) Automated text mining of the four allowed EvidenceCards to confirm the absence of biological keywords (e.g., 'senescence', 'telomere', 'mortality'). 2) Semantic similarity scoring between the query vector ('stop human aging') and evidence vectors ('stop squark', 'stop sign') to quantify the semantic collision. 3) A negative control validation to ensure that if relevant biological evidence were present, the pipeline would detect it. This approach transforms a 'null result' into a verifiable scientific finding: that the current evidence set is structurally incapable of answering the biological question due to domain mismatch.

Datasets

Source

```
```json
[
{
 "id": "EV-Q025-8a6346824f77edf2b35a2b74",
 "description": "Particle physics paper on stop squark mass splitting; used as negative control for biological content."
},
{
 "id": "EV-Q025-b0f8d69a416d10c7427609b9",
 "description": "Computer vision paper on stop sign detection; used as negative control for biological content."
},
{
 "id": "EV-Q025-7654cac739beaaaa3d5b8e33",
 "description": "Particle physics paper on LHC cross-section limits for stop particles; used as negative control."
},
{
 "id": "EV-Q025-5cebb360904beebdc6f4d5dc",
 "description": "Particle physics paper on SUSY parameter space and stop NLSP; used as negative control."
}
]
```
```

Target

Verification Report confirming zero biological relevance in source datasets and quantifying semantic distance between query and evidence.

Paper Abstract

Background: Questions regarding the cessation of human aging are often conflated with unrelated scientific domains due to lexical ambiguity (e.g., the word 'stop'). Objective: To formally verify that the provided evidence set, consisting of high-energy physics and computer vision literature, contains no valid information regarding human biological aging or mortality plateaus. Methods: We performed a systematic keyword analysis using MeSH terms and semantic embedding analysis using SciBERT to quantify the relevance of four specific evidence cards

(EV-Q025-8a6346824f77edf2b35a2b74, EV-Q025-b0f8d69a416d10c7427609b9, EV-Q025-7654cac739beaaaa3d5b8e33, EV-Q025-5cebb360904beebdc6f4d5dc) to the query 'Can we stop ourselves from aging?'. Validation Plan: The hypothesis of insufficiency will be validated if zero biological keywords are found and semantic similarity scores remain below the threshold for topical relevance. Results: pending. Conclusion: This study demonstrates the critical importance of domain-specific evidence grounding and highlights the risks of semantic collision in automated scientific question answering.

Methods

1. Keyword Extraction & Filtering: Apply a curated biomedical ontology (MeSH terms related to aging) to scan full text of all four EvidenceCards. 2. Semantic Embedding Analysis: Use pre-trained language models to generate embeddings for the query and each evidence card, calculating cosine similarity. 3. Manual Verification Protocol: Double-blind review by domain experts to classify each card's primary domain. 4. Statistical Validation: Compute the probability of false negative (missing biological content) given the search strategy.

Experiments

Baselines

```
```json
```

```
[
```

```
"Random Keyword Match: Baseline expecting high relevance due to shared token 'stop'.",
```

```
"General Domain Classifier: A standard text classifier trained on arXiv categories to predict 'physics' or 'cs.CV' vs 'q-bio'."
```

```
]
```

```
```
```

Metrics

```
```json
```

```
[
```

```
"Biological Keyword Hit Rate: Number of MeSH aging-related terms found per document (Expected: 0).",
```

```
"Semantic Similarity Score: Cosine similarity between query embedding and evidence embeddings (Expected: <0.2 for biological relevance).",
```

```
"Domain Classification Accuracy: Percentage of cards correctly classified as non-biological by the baseline classifier."
```

```
]
```

```
```
```

Ablation

Remove the word 'stop' from the query to test if any latent biological relevance exists in the evidence cards unrelated to the keyword match.

Validation Protocol

Cross-validate findings by checking if any cited references within the four evidence cards point to biological studies. If no internal citations are biological, and full-text analysis yields zero hits, the 'Insufficient Evidence' conclusion is validated.

Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

References

- ****EV-Q025-8a6346824f77edf2b35a2b74**** · arxiv · arXiv:1212.6847
 - authors: Not available · year: Not available
 - url: <https://arxiv.org/pdf/1212.6847.pdf> · doi: Not available
 - reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:5|section:page-5|paragraph:1;
content_sha256=d5fb305f1d48557cf2fe8dfdc11c83f9475a993e400900c1e56f005ac433ad33
- ****EV-Q025-b0f8d69a416d10c7427609b9**** · arxiv · arXiv:1710.03337
 - authors: Not available · year: Not available
 - url: <https://arxiv.org/pdf/1710.03337.pdf> · doi: Not available
 - reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:2|section:page-2|paragraph:1;
content_sha256=28e6aae88c6676ba0bc0767e77b7ff4a97a584f33d35694fb5be05223cf1f
- ****EV-Q025-7654cac739beaaa3d5b8e33**** · arxiv · arXiv:1401.7989
 - authors: Not available · year: Not available
 - url: <https://arxiv.org/pdf/1401.7989.pdf> · doi: Not available
 - reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:7|section:page-7|paragraph:1;
content_sha256=b461750a4625ceda97ae7c8a21274da127de83e41bac8b2972077a29b215136e
- ****EV-Q025-5cebb360904beebdc6f4d5dc**** · arxiv · arXiv:1107.2128
 - authors: Not available · year: Not available
 - url: <https://arxiv.org/pdf/1107.2128.pdf> · doi: Not available
 - reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:8|section:page-8|paragraph:1;

Reviewer Comments

- The candidate hypothesis correctly identifies 'insufficient_evidence' due to domain mismatch (particle physics/computer vision vs. biology), strictly adhering to the prohibition against hallucinating biological facts from irrelevant sources.
- Experiment design appropriately reframes the task as a verifiable metadata/semantic analysis protocol rather than a wet-lab experiment, which is the only valid approach given the null evidence set.
- Results field correctly states 'pending' and does not fabricate validation outcomes for the proposed text-mining verification.
- All cited evidence_ids exist in the provided EvidenceCards and are accurately characterized as non-biological based on quoted_text content.
- Falsifiability is maintained via the negative control logic: if any allowed evidence card contains biological keywords, the insufficiency claim is refuted.

Revision History

Reproducibility Checklist

- Verify access to full text of all four EvidenceCards via provided URLs.
- Ensure consistent version of biomedical ontology (MeSH) is used for keyword filtering.
- Document the specific embedding model and parameters used for semantic similarity calculation.
- Record inter-rater agreement scores from manual domain classification.